Corteva Agriscience

Industry: Agricultural Chemicals **Current Price (as of 19 August 2024): $54.45**

Recommendation: BUY **Target Price: $66.33**

Company Overview

COMPANY STATISTICS	
Market Cap (Billion of US$)	36.52
52-Week Range (US$)	43.22 - 58.76
Dividend Yield (%)	128.00%
Free Cash Flow Yield (Million of US$)	788.00
P/E (Current)	41.20
Return on Equity (%)	2.19%

Corteva Agriscience (CTVA) became an independent business after it spun off from DowDuPont in 2019. The business revenue flows into two major segments, seed and crop protection. For the seed segment, Corteva focuses on generating advanced germplasm and traits to enhance agricultural productivity, especially in key regions like North America, Europe, Brazil, and Argentina. The Crop Protection segment markets its products to over 125 countries globally, safeguarding crops against diseases, pests, and weeds.

By 2023, North America was responsible for half of Corteva's revenues, followed by EMEA (Europe, Middle East, and Africa) at 23%, Latin America at 19%, and Asia Pacific at 8%. The pioneer agency model supports a direct sales approach through an extensive distribution network that enables close interaction with farmers to tailor products for different regions.

AI Focus: Innovation is a deep-seated value for Corteva, as demonstrated by the $1.3 billion R&D investment in 2023, which resulted in more than 400 new products transforming agriculture worldwide. In addition to innovation, strategic partnerships have been formed, such as with Google, which has enabled them to integrate advanced AI solutions into their products, thus boosting the accuracy of their crop yield predictions.

Market Position: In the industry, Corteva holds a strong market position with a 14.89% EBITDA margin, surpassing the average of 12.9%. The firm focuses on sustainable agricultural practices that align with global trends and hence positions as one of the sector pioneers. Corteva relies on its rich heritage from DowDuPont to stay competitive in agrochemicals, as it is relatively new.

Shareholders: At 84.9%, most of Corteva's shareholders are institutional investors who own many outstanding shares. Strategic purchases, efficient operations, and a commitment to sustainability and innovation have kept the company's value high.

Investment Thesis

We rate Corteva (CTVA) as a **Buy** with a Target Price of $66.33. The following key pillars underpin our bullish stance:

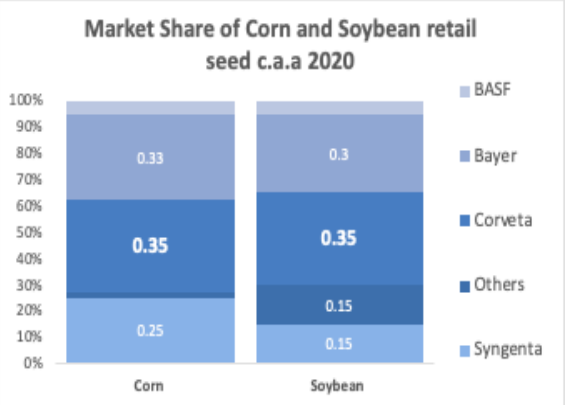
- ➔ **Pure-Play Agricultural Focus**—CTVA focuses undividedly on agricultural product offerings compared to larger but heavily diversified competitors, positioning it for superior growth. This is evident in CTVA's higher-than-industry ROA (0.97%).
- ➔ **Strategic Focus on Innovation** — CTVA has committed to investing approximately 8% of revenue in R&D by 2025, demonstrating an exemplary focus on innovation. Such strategic investments are expected to yield industry-leading ROI, doubling that of competitors, while partnerships with tech leaders such as Google to leverage AI solutions have allowed CTVA to provide instant value to customers.

Today, the Pioneer Seeds app has more than 20,000 users and assists farmers in estimating over 121,000 different crop yields more accurately and further into the future than before.

- ➔ **Operational Excellence Driving Margin** — Implementation of significant reforms and cost-saving initiatives is set to improve operating margins. Despite higher commodity costs, CTVA grew Seed Operating EBITDA by 16% and enjoyed $100mm productivity savings for Seed and Crop Protection. CTVA forecasts a total of $300mm in cost benefits in 2024 and enjoys an impressive net margin of 5.31%, showing strong profitability and cost controls.
- ➔ **Unique Routes to Market**—CTVA operates a diverse worldwide, multi-channel, multi-agency network that markets and distributes CTVA's brands to customers. The four differentiated channels include Pioneer agency, regional and retail brands, and third-party licensing. These unique routes allow CTVA to work face-to-face with farmers worldwide, to understand their needs better and more effectively market their offerings.

Market Analysis

Global Seed Market: The global seed market had an estimated value of USD 73 billion by 2024 and is likely to expand up to USD 99.94 billion by 2030, growing at a CAGR of 5.38% during the forecast period from 2024 to 2030. Corteva and Bayer held over half the U.S. retail seed sales for corn, soybeans, and cotton in the year ended on the most recent estimates in 2018–20.



North America accounted for a major share of revenue, with about $25.4B, or about one-fourth of total North American seed market revenues in 2023. Oil seeds also recorded a high growth rate among crop family markets due to expanded acreage under GM oilseed crops and rising demand for soya beans by food and feed industries.



Crop Protection Market: The global crop protection market was valued at around USD 91.4 billion in 2023 and is likely to grow at a CAGR of 4.2% between now and 2032.

The primary drivers for crop protection chemicals are pest management, disease control, and weed control. These chemicals negatively affect crop yield quality-wise and globally. Pests invade agricultural ecosystems more often, increasing the need for effective chemical solutions.

The US had the largest market share, generating $36.5 billion in revenue in that same year alone; this segment is expected to continue expanding until it catches up with other countries or surpasses them completely by 2032. Lastly, since a wide range of diseases affect different crops grown here, American demand for crop protection chemicals is expected to sustain well into the future.

Competitive Positioning for Corteva

Moat Analysis: Wide Moat

Source of Competitive Protections:

- ➔ **Extensive Investments in R&D and Innovation**:
  - ◆ Corteva spent $1.3 billion in research and development in 2023, making it one of the top spenders on R&D in the agriculture industry. This resulted in the creation of 400 new products in 2023 across seeds and crop protection. Corteva's EBITDA margin is 14.89%, meaning that these innovations have led to high profitability compared to the average industry of 12.9%. Despite being an agricultural pure-play, the company has 5900 US patents, including 11,500 international ones, which makes it a technological leader in this sector. Some of these include share gains for its Enlist E3 soybeans brand, which achieved a 65% penetration rate by year-end, leading to its rise to become USA's best-selling soybean technology in terms of volume sales, according to USDA data.
- ➔ **Integrated Product Portfolio**:
  - ◆ Corteva's integrated product portfolio contributed a total revenue of $16.9 billion in 2023, with the seed segment alone accounting for 55% of this revenue and the crop protection segment contributing 45%. This showcases a balanced and diversified product offering. The company's crop protection products are utilised by farmers in over 125 countries, offering a comprehensive suite of solutions that address local and global agricultural challenges.
- ➔ **Extensive Distribution Network**:
  - ◆ Corteva's global distribution network spans more than 130 countries, with a significant presence in North America, where it generated 46% of its 2023 revenue. Corteva's pioneer agency model further enhances this distribution network, directly connecting with farmers and providing tailored solutions and real-time insights. This direct interaction model has contributed to 10.66% YTD performance, reflecting the market's confidence in its distribution strategy.

	P/E	EV/ EBITDA	EBITDA Margin (%)	ROA	YTD Price Performance
Corteva (NYSE: CTVA)	**41.26**	**15.40**	**14.89**	**2.10%**	**10.66%**
Bayer (FRA: BAYN)	28.61	5.92	20.68	-1.05%	-21.78%
BASF SE (FRA: BAS)	1,457.83	8.43	10.58	-0.05%	-7.34%
FMC Corp. (NYSE: FMC)	5.25	27.13	10.27	11.50%	0.19%
Nutrien Ltd. (NYSE: NTR)	28.68	8.38	15.59	1.48%	-15.55%
KWS SAAT SE (FRA: KWS)	11.54	8.33	16.69	4.47%	21.91%
Nufarm Limited (ASX: NUF)	104.25	9.87	8.37	2.39%	-25.96%

Porter's Five Forces:

- ➔ **Bargaining Power of Customers – Insignificant (5 out of 5)**:
  - ◆ Corteva's pricing power is evident from its ability to maintain premium pricing for its advanced seed technologies and crop prediction products in the face of economic challenges. Despite a 6% decline in volume and a 2% unfavourable currency impact, Corteva was comfortable with increasing prices by 13% in 2023, resulting in a 5% increase in sales.
- ➔ **Intensity of Competitive Rivalry – Low to Medium(3 out of 5)**:
  - ◆ Although Corteva operates in a highly competitive industry with major MNCs such as Bayer, BASF, Syngenta (ChemChina), and FMC, and numerous regional seed companies, it maintained a strong Revenue YoY growth at 1.1% due to investments in R&D, pricing power, and a strong brand reputation.

- **Bargaining Power of Suppliers – Low (4 out of 5)**:
  - Backward Vertical Integration: Corteva engages in extensive backward vertical integration by conditioning and packaging seeds via its plants. It also diversifies and relies on third-party growers and corn manufacturers worldwide, which suggests a balanced approach. Corteva owns 99 manufacturing facilities strategically placed across North America, EMEA, Latin America, and Asia Pacific.
  - Forward Vertical Integration: Corteva's pioneer agency model exemplifies its forward integration, which enables direct sales and continuous interaction with farmers, providing real-time insights and tailored advice. This approach and regional and retail brands ensure a strong market presence and close customer connections through various distribution channels.



- **Threat of Substitutes – Low (4 out of 5)**:
  - Substitutes in the agricultural sector include organic farming practices or alternative pest control methods. However, for large-scale commercial farming, the need for high-yield, reliable seeds and effective crop protection products makes the threat of substitutes relatively low.
- **Threat of New Entrants – Insignificant (4 out of 5)**:
  - Entering the agricultural industry requires significant capital investment, regulatory approvals, and R&D capabilities, particularly in seeds and crop protection. The capital expenditure (CapEx) to sales ratio typically hovers around 6% for the agricultural inputs sector. The established presence of major players like Corteva, along with their customer loyalty and distribution networks, creates a high barrier to entry.

Revenue Drivers:

Below are the main drivers behind Corteva's (CTVA) revenue growth:

- **Robust Pricing Ability**—CTVA successfully implemented a 5% price increase across Seed offerings, leading to revenue gains and capturing value for technology. In FY23, CTVA grew Seed net sales by 5% with a 13% price increase. CTVA's dominant regional market positions enable the leveraging of brand strength and technological advancements to gain pricing power.
- **Industry-Leading Product Innovation** — CTVA's strong innovation led to attractive product offerings, driving robust demand for new products. CTVA launched more than 400 new products in 2023 while offerings such as A-Series E3 soybeans which demonstrated significant yield advantages over competitors quickly became the number one selling soybean technology in the US (60% market share gain in 5 years). CTVA's seeds enjoy considerable customer loyalty and pricing power with >65% market penetration in 2024. Based on McKinsey estimates, CTVA is well-poised to make significant revenue gains (>2.5%) in Seed and Crop Protection segments with AI integration while other innovations are expected to yield significant royalty premiums. Royalty expenses are also projected to decrease by 40% from 2020 to 2025 while royalty income triples.
- **Strategic Alignment with Sustainable Agriculture Trends** — Product development aligns with the United Nations Sustainable Development Goals (UN SDGs). By 2025, 100% of pipeline innovations will meet Sustainable Innovation Criteria, positioning CTVA as a leader in environmentally friendly agricultural solutions. This alignment enhances long-term market positioning and revenue potential in the growing sustainable agriculture sector.

Financial Analysis

Key Financials (Yahoo Finance)

Key Financials	2020	2021	2022	2023	TTM
Income Statement (Millions of US$)					
Revenue	14,217.00	15,655.00	17,455.00	17,226.00	16,901.00
Gross Profit	**5,710.00**	**6,435.00**	**7,019.00**	**7,306.00**	**7,421.00**
R&D Expense	**1,142.00**	**1,187.00**	**1,216.00**	**1,337.00**	**1,381.00**
Operating Expense	4,860.00	5,118.00	5,091.00	5,196.00	5,287.00
Operating Income	**843.00**	**1,317.00**	**1,928.00**	**2,110.00**	**2,134.00**
EBITDA	1,897.00	3,619.00	2,728.00	2,537.00	2,517.00
Net Income	681.00	1,759.00	1,147.00	735.00	898.00
Balance Sheet (Millions of US$)					
Total Assets	42,649.00	42,344.00	42,618.00	42,996.00	-
Total liabilities	17,586.00	16,721.00	17,077.00	17,717.00	-
Total Equity	**25,063.00**	**25,623.00**	**25,541.00**	**25,279.00**	-
Total Debt	**1,105.00**	**1,117.00**	**1,307.00**	**2,489.00**	-
Operating Cash Flow	2,064.00	2,727.00	872.00	1,769.00	1,395.00
Per Share Data (US$)					
EPS (Diluted)	**0.91**	**2.37**	**1.58**	**1.03**	**1.30**
Dividend per share	**0.52**	**0.54**	**0.58**	**0.62**	**0.68**

Key Performance Indicators (KPIs):

- **Revenue**: The company's annual revenue increased, from **$14.2 billion in 2020** to **$17.2 billion in 2023**. This shows a growth in its business by about **21.1% over the last 4 years.**
- **Net Income**: The company's net income rose from **$681 million in 2020** to an estimated **$1.76 billion in 2021**. However, due to the restructuring between 2022 and 2023, its net revenue dropped by a small margin as seen in the table above. Yet, this does not define the future potential as it is seen to be rising again for the trailing 12-month period.
- **Earnings Per Share (EPS) diluted**: EPS (diluted) has generally increased from 1.03, in 2023 to 1.30, in 2024, indicating the increased **profitability of the stock**. An increasing EPS, suggests a drop in the P/E ratio (currently at 41.3), suggesting that the stock is at a bargain cost currently.
- **Return on Equity (ROE):** The ROE of Corteva is seen to be moderate to low between a range of **2% to 6%** between **2020 to 2024**. Although the ROE is lower than the **industry average of 16% (Simply Wall St, Yahoo Finance, 10 Apr 2023)**, they have had a **payout ratio of about 30%**. This insinuates a slow yet gradual payout and consistency and stability in earnings in the long run. This also insinuates that their earnings may be pumped back into the company as reinvestments, projecting a future growth potential.

Balance Sheet Analysis:

- **Total Assets**: Corteva's total asset value has been stable over the last 5 years.
  - 2020: Approximately **$42.6 billion**
  - 2023: Approximately **$43 billion**
- **Total Debt**: The company's debt levels have been managed prudently, suggesting a well-balanced capital structure. The debt-to-equity ratio of Corteva has been 10.9%, which is seen to be satisfactory, and utilising its debt to drive growth and expansion.
  - 2020: Total debt was around **$1.1 billion**
  - 2023: Total debt has been relatively stable, around **$2.5 billion**

Performance of the stock price:

➔ **Stock Price**: Corteva's stock has been volatile but has had an up-trend. The stock price increased from around **$27 per share** in 2019 to around **$54 per share in mid-2024**. The value of its dividends per share has been consistently rising at a rate of **6% per annum** since 2020, which shows stability in its financials. The stock price is currently stabilising around **$54 per share.** With an increasing EPS and stable revenue growth, the stock value is a bargain at this price and is expected to rise.



Summary:

Corteva Agriscience has prospered over the past years evident through its increasing revenue and net income despite restructuring between 2022 and 2023. The increase in stock price and dividend per share value shows this company's profitability. Their successful implementation of AI and product innovation is the key to its success. Its embedded focus on product innovation and improved productivity portrays a bright future for this company. Their operating **EBITDA** has shown a **10% increase since the 2Q of 2023**, proving the profitability of this company.

($ in millions, except EPS)	2Q 2023	2Q 2024	Change
Net Sales	$6,045	$6,112	+1%
GAAP Income (Loss) from Continuing Operations After Income Taxes	$880	$1,056	+20%
Operating EBITDA[(1)]	$1,746	$1,917	+10%
Operating EBITDA Margin[(1)]	28.9%	31.4%	+248 bps
GAAP EPS from Continuing Operations	$1.23	$1.51	+23%
Operating EPS[(1)]	$1.60	$1.83	+14%

Valuation

For our base case, our Discounted Cash Flow Model implies a fair value of **US$66.33** as of 15 August 2024, with a projected enterprise value of approximately US$4.6 bil, representing a **25.7% upside** from the current share price of US$52.76. Our DCF model utilises a terminal growth rate of 2.5%, in line with the projected GDP growth of the US economy and 126 basis points below the projected CAGR of the agricultural market from 2024 through 2029. The calculations for the Weighted Average Cost of Capital shall be discussed in brief later. Our downside share price valuation of US$56.10 represents a

Model Breakdown:

Corteva's revenue can be disaggregated into its crop protection and seeds segment. At the time of this report, **seeds constitute around 55% of revenues** while **crop protection** makes up the **remaining 45%**. growth/change in net sales for **each** segment can be broken down into:

➔ Net sales change from volume
➔ Net sales change from price/product mix
➔ Net sales change from foreign exchange implications

Our approach was to first derive the net sales change(%) stemming from each segment by modelling the net sales change contributed by each factor, weighted by region. Finally, a net sales growth rate is generated for each year by taking the weighted sum of the net sales change of each segment according to the aforementioned sales segmentation. We have also sensitised Corteva's SG&A which is projected by management to experience

at least US$100 mil of run-rate cost savings starting in 2025 due to Corteva's extensive restructuring. Relatedly, we have modelled its restructuring-related charges and expenses to taper off from 2025 onwards. 2024 remains a transitory period as the last portion of major restructuring, amounting to US$410 mil, associated with the Crop Protection Operations Strategy Restructuring program is set to occur this year. With the ongoing restructuring and residual destocking damping sales volume in 2024, we only observe Corteva's full growth trajectory starting in 2025. We have also noted that the substantial price increases that have driven growth in 2023 and 2022 are linked to inflation and will thus taper off in tandem with projected inflation numbers as inflationary pressures subside.

	UPSIDE	BASE	DOWNSIDE
Implied Share Value	71.67	66.33	56.1
5 Year Revenue CAGR	5.3%	3.5%	1.1%
Year 5 EBITDA	3768	3421	2116
Year 5 EBITDA Margin	16.5%	16.5%	14.5%
Year 5 FCFE, undiscounted	2831	2620	2196
Total Cost Savings by Year 5 (US$mil)	1700	1500	500
Net Sales Growth CP	Stabilising at 6.8% 2025 onwards	Stabilising at 4.0% from 2025 onwards	Stabilising at 2.6% 2025 onwards
Net Sales Growth Seed	Stabilises at 6.6% from 2025 onwards	Mid 4% range from 2025 onwards	Stabilising at 3.1% 2025 onwards
WACC	7-9.5%	7-9.5%	7-9.5%
Terminal Growth Rate	2.5%	2.5%	2.5%

Discussion:

The key model features are as given. The cost was calculated using the CAPM, with the beta being benchmarked against the Invesco DB Agriculture Fund and the 10-year US treasury yield being used as the risk-free rate. The cost of debt was taken to be the weighted average rate of its debt. The derivation of the base case share price is as shown. Cashflow timing is assumed to be at the end of the year.

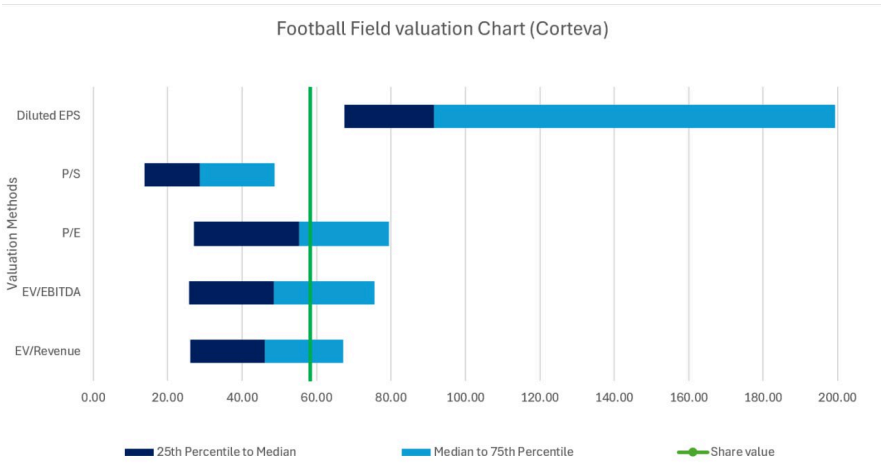
Build up to FCF						
	Valuation Date					
Fiscal Year End Date	15/8/2024	31/12/2024	31/12/2025	31/12/2026	31/12/2027	31/12/2028
		FY2024E	FY2025E	FY2026E	FY2027E	FY2028E
Net Cash Provided by Operations		2138	2587	2669	2986	3343
CapEx		-611	-640	-667	-695	-723
Free Cash Flow to Equity		1528	1947	2002	2291	2620
Partial Period Adjustment		38%	100%	100%	100%	100%
FCFE Adjusted for Partial Periods		577	1947	2002	2291	2620
Years for Discounting		0.4	1.4	2.4	3.4	4.4
WACC		7.60%				
Discounted Cashflows		561	1760	1682	1789	1901
Net Debt		-253				
No. Outsatnding Shares as of Valuation Date		695,900,000				
Terminal Value Calculation						
Terminal Growth Rate	2.5%					
Terminal Value	52657					
Discounted Terminal Value	38211					
Stock Price Calculation						
Enterprise Value	45904					
Equity Value	46157					
Implied Share Price	66.33					

On the right is a sensitization of the WACC (column) and terminal growth rate (row) for our base case. Concerning the base scenario, even though the company's top line grew by a **CAGR of 3.5%**, its EBITDA margin saw a strong expansion amounting to **220 basis points from 14.3% in 2023 to 16.5% in the final year of projection (2025)**. We predict Corteva's restructuring to be a key value and profitability driver, producing a base case projection of **US$1500 mil in cost savings by 2025**. Improved profitability is also driven in part by the fact that restructuring charges are expected to taper down past 2024. In the same vein of thinking, we project net

		WACC					
		7%	7.60%	8%	8.50%	9%	9.50%
Terminal Growth	2%	68.70	61.18	57.00	52.51	48.66	45.32
	2.50%	75.36	66.33	61.40	56.17	51.75	47.96
	3%	83.68	72.59	66.68	60.50	55.36	51.00
	3.50%	94.38	80.39	73.13	65.70	59.62	54.56
	4%	108.65	90.34	81.20	72.05	64.74	58.76
	4.50%	128.62	103.52	91.56	79.99	70.99	63.79

sales growth to stabilise and return to a growth trajectory from 2025 onwards as the company starts to reap the dividends of its restructuring and develop better operational execution concerning its star products.

We have also determined that Corteva can sustain its strong stock buyback policy and generous dividend payouts given it will **resume growth in 2025.** We have projected that the company can remain cash flow positive while growing its dividends in tandem with revenue from **US$439 mil in 2023 to US$457 mil and execute its 2022 Share Buyback Program**, incurring an estimated US$350mil yearly for the projection period. This is a strong value add for shareholders.



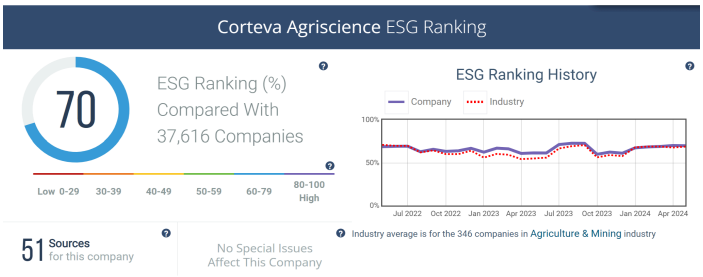
The above chart displays our target price in the context of the range of implied share values derived from public company comparables. Our base case target price sits within a reasonable range, slightly above the median implied by most multiples, indicating that only some upside has been priced in. There is still room for growth. It should also be noted that our downside price target of US$56.10 is higher than the stock's current price, reflecting our confidence in Corteva's recovery. The uptick in the company's performance in the first two quarters of 2024 substantiates this.

Investment Risks

- **Market Competition:**
  - **High Price fluctuations:** The price fluctuations of agricultural commodities like seeds, corn, and soybeans can impact farmers' purchasing decisions and, consequently, Corteva's revenues. This risk is further aggravated by its high competition.
  - **Market Competition:** Bayer, Syngenta, and BASF are other large firms competing with Corteva. Intense competition can pressure margins and impact market share.
  - **Technological Advancements:** Rapid changes in agricultural technology and innovation become the cornerstone for the company to beat its market competition, altering its growth potential.
- **Inter and Intra National Regulations:**
  - **Regulatory Changes:** The agricultural industry is highly government-regulated, and changes in regulations, related to the agro-chemical industry, can affect Corteva's operations and availability of products.
  - **Trade Policies:** Trade restrictions and tariffs can impact global trade affecting Corteva's international sales. While having a market in over 125 countries, restrictions or tariffs may largely affect the demand for Corteva's products.
- **Natural and Environmental Risks:**
  - **Climate Change:** Changes in climate conditions and natural disasters can affect crop yields and demand for agricultural products. Rising temperatures and changing climate conditions may be critical for the demand for agricultural commodities.
  - **Sustainability Pressure:** Greater attention and awareness about sustainability may require Corteva to invest in new technologies and practices to make their processes and products more sustainable which would be discussed in the section below, potentially impacting its profitability.

ESG Considerations

Corteva Agriscience is a high performer in terms of its ESG ratings. Corteva ranks in the **70% percentile in 37,616 companies**, followed by the CSRHub. Corteva maintains a higher end of the spectrum of the ESG rating. This shows the company's high potential by effectively managing environmental, social, and government risks.



Climate:

Corteva Agriscience is committed to **reducing its intensity reduction target** (metric that sets an organisation's relative emissions to a variable) by **65%** by 2030. This metric is **consistent with the 1.5°C temperature increase** above pre-industrial levels as agreed during the **Paris Agreement** on 12 December 2015.

- ➔ **Implementing energy audits and optimizations:** Corteva is invested in cutting emissions, by auditing its electricity and gas use. They have identified multiple projects that can save an estimated 10,000 **tons of GHG** (Greenhouse Gas) emissions each year.
  - ◆ Harbor Beach, Michigan, US → Electricity and gas audits have identified 10-15% possible energy savings.
- ➔ **Increasing renewable energy use:** Corteva is selecting sites for solar panel installations and working with local partners to provide power directly to **cut 25,000 tons** of carbon dioxide emissions by 2030.
- ➔ **Purchasing Renewable Energy Certificates:** Corteva is also keen on purchasing high-quality Renewable Energy Certificates (RECs) to reduce the carbon footprint.

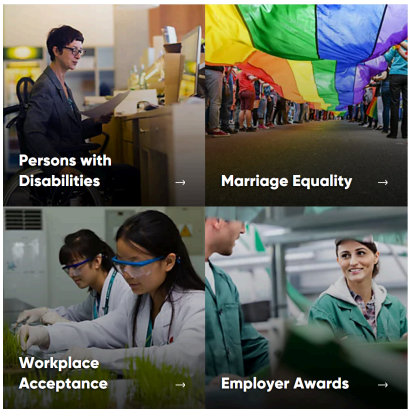


Sustainability:

- ➔ **OMRI Listed:** The Organic Materials Review Institute (OMRI) provides an independent review of products, during their production, handling, and processing phases. OMRI provides comply with the USDA National Organic Standards **(OMRI Home Page).**
- ➔ **Product Development:** New products must meet the baseline requirements to deliver notable sustainability advantages while maintaining standards and consistency compared to other products in the market today.
- ➔ **Research & Development:** Corteva capitalises on cutting-edge technology to integrate sustainable practices into its operations. With this, they aim to produce new products with high biodiversity advantages. Their **R&D expenses** for the 12 months ending on **March 31, 2024, is $1.353 Billion, a 7.04% increase** from their previous year.

Corporate Governance:

- ➔ **Inclusion, Diversity, and Equity:** Corteva is embracing inclusion and diversity by retaining top talents, and enriching the lives of their employees. They are focused on **3 key pillars; fostering an internal culture of belonging, enhancing diversity, and external impact through their work**. For example:
  - ◆ **6% year-over-year increase** in women in top management positions.
  - ◆ Between 2011 and 2022, they **empowered 600,000 farmers** in Tanzania by **increasing** their **productivity** by **300%**. There were also multiple projects in Myanmar, Africa, and Southern America.
- ➔ **Standards and Guidelines of Business Conduct:** Corteva follows stringent safety protocols to deliver food reliably and



safely. They commit to openness to increase transparency of their processes.

➔ **Ethics and compliance:** The company has established programs and training to ensure adherence to regulatory requirements and policies including anti-corruption measures.

Socials:

➔ **Improved Process Incident Severity Rate:** The governance and compliance committee of the board of directors constantly tracks review metrics to track performances to review and improve, as the safety of their employees is a priority. Their exorbitant safety standards are evident in their safety performance statistics provided in their **2022 Sustainability & ESG Report**.

- **Responsible care program:** This initiative aims to improve the health and safety of the workers. This includes emergency planning requirements. The Responsible Care Management System (RCMS) certification is mandatory by an auditor which shows their commitment to improving their health, safety, and environmental (HSE) performance **(Corteva Agriscience 2022 Sustainability & ESG Report, Page 45)**.

Safety Performance	Unit	FY 2020	FY 2021	FY 2022
Work-related Fatalities - Employees	Number	0	0	0
Work-related Fatality Rate for Direct Employees	Fatalities per 100 employees	0	0	0
Work-related Fatalities - Contractor	Number	0	0	0
Work-related Fatality Rate for Contract Employees	Fatalities per 100 contractors	0	0	0
Lost-Time Injury Frequency Rate (LTIFR)	Number per million hours worked	0.51	0.37	0.58
Process Safety Events: Tier 1	Number per million hours worked	0.05	0.00	0.05
Total Recordable Incidents Counts (TRIR)	Injuries per 100 employees	0.35	0.36	0.39

- **Management system Certification & third-party assurance:** Employees and contractors are **100% covered for occupational health and safety incidents** while working.
- **Health and safety assurance:** Corteva engages with SGS United Kingdom Ltd to **provide independent assurance to the wellbeing of their employees**. SGS is a reputable company that uses procedures and systems that provide accurate data with transparency and consistency. They comply with the **internationally recognized standards and guidance**. They confirmed Corteva's health and safety protocols are of the highest standards with no evidence that proves otherwise. This is evident by Corteva having 0 fatalities and having a low injury frequency rate.

➔ **Community Impact:** Corteva emphasises community impact through various initiatives and programs. Their community-focused efforts are a key part of their corporate social responsibility.

- **Education and Scholarships:** Corteva partners with educational institutes to promote careers in agriculture and science by offering scholarships to students. They have offered **25 scholarships**, by partnering with the Des Moines Area Community as part of a Pilot Programme on June 5, 2019. This was done as part of their efforts to renew rural areas and in the talented students to solve future problems.



- **Volunteerism and social work:** Corteva Agriscience engages in various social work initiatives. These initiatives create a positive impact on communities.
  - **Access to Nutritious Food:** Corteva collaborated with The Akshaya Patra Foundation in India to address the issue of food insecurity by facilitating the delivery of warm and nutritious meals benefitting almost **100 schools, 400,000 meals per year.**
  - **Fighting hunger in the local community:** Corteva has produced and **donated over 100,000 pounds of local food pantries** since 2014, in Indiana.

Conclusion

Corteva is an excellent investment case with its focus on agricultural innovation and excellence in operations, which allows it to compete effectively against other competitors that have also embraced artificial intelligence technologies. Through strategic partnerships with AI leaders like Google, Corteva is well positioned to take advantage of growth patterns in sustainable agriculture because they enjoy strong pricing power and a diversified product portfolio. Despite competition from other market players and some sector-wide challenges, Corteva's robust financials expect positive growth prospects, and cost management efficiency make it an even more attractive investment. The company's efforts towards community involvement and sustainable living also indicate how much potential for growth exists within a firm like this one. According to our valuation model, we ought to **BUY** the stock with an expected target price of **$66.33 per share**, implying a **25.4% upside potential.**

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Appendix A (Calculations for Football Field Valuation)

Comparable Companies Analysis (US Agrochemical Industry)

Company	Market Data (TTM)					Financials			Valuation				
	Share Price	Shares Outstanding	Equity Value	Net Debt	Enterprise Value	Revenue	EBITDA	Net Income	EV/Revenue	EV/EBITDA	P/E	P/S	Diluted EPS
Corteva Agriscience	53.80	703	37,240.00	2,489.00	40,150.00	16,901.00	2,517.00	898.00	2.38	15.95	41.47	2.24	1.30
DuPont de Nemours	80.18	544	33,470.00	5,339.00	39,090.00	12,058.00	1,930.00	664.00	3.24	20.25	50.41	2.87	0.83
Mosaic Co	27.72	324	8,830.00	3,412.60	12,940.00	12,193.80	1,468.90	244.80	1.06	8.81	36.07	0.74	0.75
FMC Corporation	63.20	125	7,890.00	3,665.20	11,600.00	4,084.40	419.20	1,384.40	2.84	27.67	5.70	1.94	11.06
Bayer AG	32.17	982	31,428.74	39,414.45	67,049.46	51,972.71	10,751.30	- 1,396.59	1.29	6.24	- 22.50	0.66	- 1.42
High									3.2x	27.7x	50.4x	2.9x	11.1x
75th Percentile									2.9x	22.1x	39.7x	2.2x	3.4x
Average									**2.1x**	**15.7x**	**17.4x**	**1.6x**	**2.8x**
Median									**2.1x**	**14.5x**	**20.9x**	**1.3x**	**0.8x**
25th Percentile									1.2x	8.2x	-1.4x	0.7x	0.2x
Low									1.1x	6.2x	-22.5x	0.7x	-1.4x

Corteva Valuation	EV/Revenue (Min)	EV/Revenue (25th)	EV/Revenue (Median)	EV/Revenue (75th)	EV/Revenue (Max)
Implied Enterprise Value	17,935.26	20,836.67	34,902	49,698	54,790
Total Debt	2,489	2,489	2,489	2,489	2,489
Implied Market Value	15,446.26	18,347.67	32,412.95	47,208.62	52,301.19
Shares Outstanding	703	703	703	703	703
Implied Value Per Share	**21.97**	**26.10**	**46.11**	**67.15**	**74.40**

Corteva Valuation	EV/EBITDA (Min)	EV/EBITDA (25th)	EV/EBITDA (Median)	EV/EBITDA (75th)	EV/EBITDA (Max)
Implied Enterprise Value	15,697.0	20,554.04	36,576	55,647	69,650
Total Debt	2,489	2,489	2,489	2,489	2,489
Implied Market Value	13,208.0	18,065.0	34,087.0	53,157.7	67,160.8
Shares Outstanding	703	703	703	703	703
Implied Value Per Share	**18.79**	**25.70**	**48.49**	**75.62**	**95.53**

Corteva Valuation	P/E (Min)	P/E (25th)	P/E (Median)	P/E (75th)	P/E (Max)
Implied Market Value	- 20,208.51	- 1,213.70	18,754	35,610	45,265
Shares Outstanding	703	703	703	703	703
Implied Value Per Share	**- 28.75**	**- 1.73**	**26.68**	**50.65**	**64.39**

Corteva Valuation	P/S (Min)	P/S (25th)	P/S (Median)	P/S (75th)	P/S (Max)
Implied Enterprise Value	11,154.66	12,168.72	22,647	36,717	48,506
Total Debt	2,489.00	2,489.00	2,489.00	2,489.00	2,489.00
Implied Market Value	8,665.66	9,679.72	20,158.34	34,228.42	46,016.87
Shares Outstanding	703	703	703	703	703
Implied Value Per Share	**12.33**	**13.77**	**28.67**	**48.69**	**65.46**

Corteva Valuation	Diluted EPS (Min)	Diluted EPS (25th)	Diluted EPS (Median)	Diluted EPS (75th)	Diluted EPS (Max)
Implied Value Per Share	**- 58.89**	**8.61**	**32.76**	**140.48**	**458.66**